

Internet of UAVs to Automate Search and Rescue Missions in Post-Disaster for Smart Cities

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Abstract— In natural disasters, the emergency rescue system is of utmost importance for the smart city development, but the search and rescue (SAR) leveraging unmanned aerial vehicles (UAVs) is under-explored. This study has established an Internet of UAVs architecture motivated by search and rescue activities, focusing on the path planning problem of UAV in three-dimensional urban disaster scenes. A mathematical model for UAV-based SAR missions has been built, aiming to use UAV platforms equipped with life-detection radars to search for survivors inside buildings. Our aim is to search for the most survivors in the shortest amount of time while ensuring coverage of the entire search area. Based on the mathematical model for SAR activities, the improved Multi-Verse Optimizer (IMVO) is used to solve the path of UAV. Finally, simulations were conducted in a photorealistic urban scenario, demonstrating the paths generated by the proposed method. The results indicate the potential of the generated path in terms of search time and the number of survivors discovered.

Index Terms—UAV, Search and Rescue, Path Planning, Multi-Verse Optimizer, Smart Cities

I. INTRODUCTION

From 2010 to 2020, natural disasters (earthquakes, floods, typhoons, etc.) have caused an average of 37,400 deaths and economic losses of 187 billion US dollars per year worldwide [1]. Therefore, investment in disaster risk management (DRM) is increasing globally to enhance the sustainability and resilience of cities and communities [2]. Meanwhile, sustainability and resilience are also important indicators for evaluating smart cities [3]. With the rapid progress of control, communication, and artificial intelligence technology, the application of UAVs in the field of DRM has expanded significantly, such as emergency communication based on UAVs [4] and emergency logistics based on UAVs [5]. In these application scenarios, a challenging task is to deploy

This work was supported in part by the Laboratory for Artificial Intelligence in Design, Hong Kong (Project Code: RP2-2) under the InnoHK Research Clusters, Hong Kong Special Administrative Region Government, and in part by Hong Kong Polytechnic University under Grant G-UARK.

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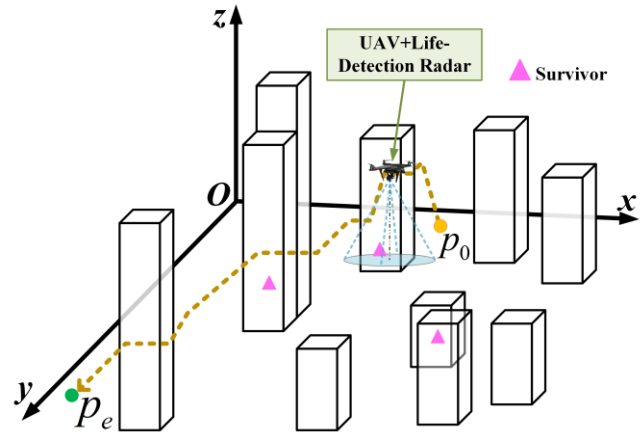


Figure 1. Schematic diagram of UAV-based search and rescue missions.

UAVs for search and rescue missions to locate survivors in post-disaster.

The goal of SAR missions is to complete the search of the entire target area in the shortest possible time. The existing solutions for UAV-based SAR tasks mainly include the use of UAV mobile networks to locate intelligent devices, and the installation of cameras on UAVs to search for survivors in exposed environments [6]. One representative framework among these solutions is the Internet of UAVs. Based on this, in view of major natural disasters such as earthquakes, typhoons and floods in which people and buildings are vulnerable, we studied the SAR path planning problem based on Internet of UAVs in smart cities. Our motivation is to use UAVs to search for survivors inside the building. Due to life-detection radars can penetrate walls to detect survivors inside buildings, this study aims to use UAVs equipped with life-detection radars to locate survivors in post-disaster. Fig. 1 shows a schematic of an autonomous search and rescue mission using the UAV.

The research purpose of this study can be determined to generate a high-quality path for UAVs equipped with life-detection radars. This problem involves several major challenges. This objective presents several significant challenges. First, as the initial 72 hours following a disaster are critical for SAR operations, it is essential to minimize both the time required for the planner to generate the path and the time needed for the UAV to execute the task. Additionally, when conducting UAV activities in urban disaster environments, the operational cost of UAVs must also be considered to extend

the effective flight time and expand the coverage of search and rescue operations.

- Based on the background of natural disasters such as earthquakes and typhoons, we built a UAV trajectory planning system for searching for and rescuing vulnerable communities, in order to have a rapid response and human-centered SAR operations.
- A mathematical model considering multi-objective of UAV-based SAR operations equipped with life-detection radar is established to search for the most survivors in the shortest amount of time while ensuring coverage of the entire search area.
- Several cutting-edge algorithms widely used in UAV path planning are used in UAV-based search and rescue experiments, and the effectiveness of the proposed methods is verified.

The rest of this paper is organized as follows. In Section II, the related research is summarized and analyzed; In Section III, the corresponding multi-objective mathematical model is established; In Section IV, The IMVO algorithm is introduced; In Section V, simulation experiments are carried out; Finally, the conclusions are given in Section VI.

II. RELATED WORK

The work related to UAV-based DRM is constantly increasing, covering various directions such as emergency communication, emergency logistics, post disaster monitoring tasks, and SAR tasks [4], [8]. Meanwhile, numerous studies have demonstrated that the hardware foundation for UAV-based DRM already exists, and Erdos et al. developed a real UAV for performing SAR tasks [9]. Macias et al. aimed to minimize the flight time of UAVs and ensure fair distribution of support materials. They combined drone fleet scheduling and drone path planning problems to study the application of UAVs in humanitarian logistics systems [10]. References [11]-[12] focused on post-disaster monitoring tasks using drones, where drone trajectories need to traverse multiple nodes and hover during post-disaster monitoring tasks. Consequently, many studies have transformed it into a variant of Vehicle Routing Problem (VRP), aiming to generate the shortest path for drones to access all nodes [11]-[13].

The difference between the above studies and those focused on UAV-based SAR missions is that, due to the maximum detection range limitation of life-detection equipment, UAVs usually need to fly at lower altitudes, which requires consideration of obstacles in the city when generating UAV flight paths [14]. Horyna et al. developed a multi-drone path planning model for searching for potential survivors in desert environments [15]. Similar literature [16] also uses a set of drones equipped with cameras for ASR activities. When generating the path of a drone, references [15]-[16] first divide the entire search area into a group of cells, and then use a metaheuristic algorithm to generate continuous paths for a group of UAVs. However, the above two studies only investigated the path planning problem of UAVs in a two-dimensional (2D) environment, without considering the risk of path collision caused by buildings.

Unlike the methods of generating paths based on discrete maps mentioned above, some path planning methods based on continuous space are also widely used, such as RRT-based [17] and spline curve-based path planning methods [18]. Based on the background of disaster emergency response, work [19] studied drone path planning, which has specific reference significance. However, it should be pointed out that it only involves drone path planning in mountainous environments and does not study SAR path planning in complex three-dimensional urban environments. Reference [20] studied the task allocation and path planning problem of UAVs in a three-dimensional urban environment, and established a mathematical model to minimize the path length.

Unlike the traditional RRT algorithm, the quality of UAV trajectories generated using metaheuristic algorithms is usually better [21]-[22]. For example, in reference [20], an evolutionary algorithm was designed and successfully applied to trajectory planning of UAV clusters in restricted environments. In addition, some studies often mix two metaheuristic algorithms to design a more efficient hybrid metaheuristic method. Compared with previous algorithms, these improved algorithms usually have better performance in terms of convergence speed and accuracy when solving UAV trajectory planning problems [23]. However, in the process of UAV aircraft planning, if only metaheuristic algorithms are used, the calculation process will become lengthy. Therefore, some studies attempt to combine traditional RRT, A* algorithm, and metaheuristic algorithm with the aim of obtaining satisfactory paths in a shorter time [24]. Similar examples include references [25]-[26]. In reference [25], an improved group search (IGSO) algorithm was designed to further optimize the initial path generated based on the RRT algorithm. Meanwhile, reference [26] designed an IMVO algorithm to optimize the initial path generated based on traditional algorithms.

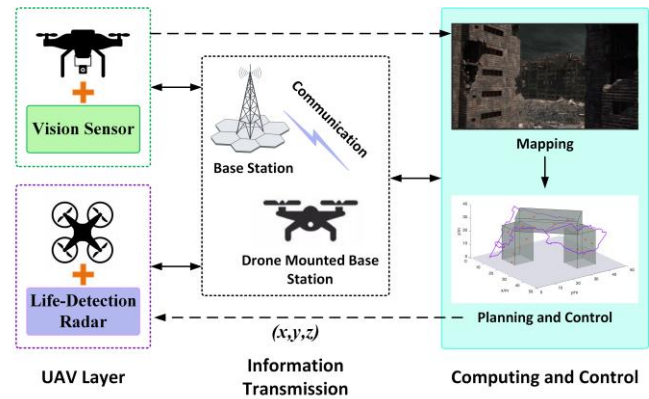


Figure 2. The Internet of UAVs architecture for SAR activities.

III. MODEL ESTABLISHMENT

After the occurrence of a natural disaster, the SAR operations should be automated and optimized by means of UAVs in order to save as many lives as possible in the first 72 hours. Fig. 2 shows an autonomous SAR framework based on Internet of UAVs, which includes two types of drones. One is a UAV equipped with visual sensors, primarily used to collect information and generate maps of disaster areas. The other type is a UAV equipped with life-detection radar, which

searches for survivors in the disaster area based on existing maps and generated trajectories. This study focuses on the path planning problem of UAVs equipped with life-detection radars.

Considering that the survival probability of trapped targets will decrease with the rescue time delay, the UAV-based SAR operations needs to consider the urgency of the search and rescue activities and shorten such time as much as possible. On this basis, in order to ensure the effectiveness of SAR activities within the specified time, the range of UAVs must cover the entire disaster area. Therefore, we stipulate that the search and rescue coverage must fully cover the affected area within a certain time threshold. In view of the demand of UAV-based SAR operations on the 3D environments, we establish a multi-target mathematical model with the objective of minimizing the SAR time, and maximizing the number of trapped targets detected by the radar life detectors.

The work [15] established a UAV path planning model for natural disaster management, but this study only considered the flight path in mountainous environments, instead of the urban environments; Moreover, it did not consider the constraints of SAR equipment carried by the UAVs, and only planned the shortest flight path between the starting and ending point. Differing from the literature [15, 20], we build a 3D environment model with 3D buildings in the urban disaster areas, focusing on the UAV track planning for performing search and rescue missions in the ground disaster area. The mathematical model is described in the following.

In this study, we assume a UAV equipped with a life detection radar that flies outside the building while detecting survivors inside. Fig. 1 shows the process of an UAV equipped with a radar life detector performing SAR missions. The maximum detection distance of the radar life detector is $R_{\max}^{Det}$, the maximum detection angle is $\theta_{\max}^{Det}$, and the single detection time of the system is T_{Det} . The flight path of UAV is represented by the set of flight path points R of UAV, $R = \{p_0, p_1, \dots, p_e\}$ and $p_r \in R$, where p_r is the r -th flight track point of UAV, $\forall r \in \{0, 1, \dots, e\}$; p_0 and p_e are the start and end points of the flight path of the UAV, respectively. In any track segment $p_{r-1} \rightarrow p_r$ of UAV, and the track segment vector is (x_r, y_r, z_r) . For the ease of calculation, it is assumed that the UAVs fly at a constant speed in each track segment. Therefore, the calculation of flight time T_r of the UAV in track segment $p_{r-1} \rightarrow p_r$ is as follows.

When $z_r \geq 0$,

$$T_r = \max\{\sqrt{(x_r^2 + y_r^2)}/V_L, z_r/V_R\} \quad (1)$$

When $z_r < 0$,

$$T_r = \max\{\sqrt{(x_r^2 + y_r^2)}/V_L, |z_r|/V_D\} \quad (2)$$

where V_L, V_R and V_D are the normal, rising and falling speed of the UAV equipped with radar life detector, respectively, usually 0.8 times of the maximum, rising and falling speed [25]. The time T for UAV to perform SAR missions is composed of the flight time of the UAV and the detection time

of the radar life detector. The calculation formula is thus presented as follows.

$$T_i = \sum_{r \in R} T_r + (\lceil (e-1)/3 \rceil + 1) \times T_{Det} \quad (3)$$

where e is the total number of flight track points of the UAV. The UAV hovers once every three track points, and the life-detector radar is activated for detection.

Therefore, this study aims to minimize the time required for UAV-based SAR operations in disaster environments and maximize the number of survivors detected by the radar life detectors. The objective function is established as follows

$$\min f = T_i \times (1 / \sum_{q \in Q} \kappa_q) \quad (4)$$

$$\kappa_q \in \{0, 1\}, \quad q \in Q \quad (5)$$

where, κ_q indicates whether the q -th trapped target can be detected by the radar life detector, and Q is the set of trapped targets, $Q = \{1, 2, \dots, q\}$.

Further, according to the physical performance of the UAVs and work condition of the life-detectors radar, the model constraints are constructed as follows.

$$\sqrt{(x_r^2 + y_r^2)}/T_r \leq V_{L\max} \quad (6)$$

$$\forall z_r \geq 0, \exists z_r/T_r \leq V_{R\max} \quad (7)$$

$$\forall z_r < 0, \exists |z_r|/T_r \leq V_{D\max} \quad (8)$$

$$T_i \leq \min\{T_{\max}, \Gamma_1\} \quad (9)$$

$$|z_r|/\|\vec{B}_r\| \leq \tan(\varphi_{\max}) \quad (10)$$

$$\cos(\phi_{\max}) \leq \cos(\vec{B}_{r-1}^T, \vec{B}_r^T) \quad (11)$$

$$\sqrt{x_{i(s),r}^2 + y_{i(s),r}^2 + z_{i(s),r}^2} \leq L_{\min} \quad (12)$$

$$\sqrt{(x_{gr}^p - x_q^p)^2 + (y_{gr}^p - y_q^p)^2 + (z_{gr}^p - z_q^p)^2} \leq R_{\max}^{Det} \quad (13)$$

$$\cos(\theta_{\max}^{Det}/2 + \varphi_{\max}) \leq |\cos(\vec{n}, \vec{O}_q \vec{p}_{gr})| \leq 1 \quad (14)$$

$$\forall m \in M \quad \exists p_{mS}, p_{mE} \in P \quad (15)$$

Formulas (6), (7) and (8) are the maximum speed constraints of normal, rise and fall of UAV during flight; Formula (9) limits the endurance time of the UAV and ensures that the UAV can complete the SAR mission in the disaster environment within the specific time threshold Γ_1 ; Formula (10) limits the maximum tilt angle $\varphi_{\max}$ of UAV, where, $\vec{B}_r = (x_r, y_r)^T$; Formula (11) limits the maximum turning angle $\phi_{\max}$ of UAV; Formula (12) is the minimum step constraint of UAV, and $L_{\min}$ is the minimum direct flight distance of UAV.

Formula (13) limits the maximum detection range of the radar life detectors, where (x_{gr}, y_{gr}, z_{gr}) are the coordinates of the hovering point p_{gr} of the UAV, and $p_{gr} \in P$, (x_q, y_q, z_q) are the centroid position coordinates of the q -th trapped target; Formula (14) is the maximum detection angle constraint of the radar life detector. During hovering of the search and rescue

UAVs to detect trapped targets, the attitude of the UAV is inclined to further expand the searching coverage. Wherein, vector $\vec{n}$ is a vector perpendicular to the ground, and $O_q = (x_q, y_q, z_q), \forall q \in Q$. When the UAV hovers at a certain hovering point, when the relative positions of the trapped targets and the UAVs meet formula (13) and formula (14), $\kappa_q = 1$; Otherwise $\kappa_q = 0$.

Formula (15) ensures that UAV can fully cover all buildings in the urban disaster area, where M is the collection of buildings in the urban disaster area, $M = \{1, 2, \dots, m\}$. p_{ms} and p_{me} are the starting and end points of the track segment where the UAV perform search and rescue missions on the building m . When the UAV passes through p_{ms} and p_{me} in the flight process, it is deemed that the UAV covers the building m for SAR operations.

IV. THE IMPROVED MULTI-VERSE OPTIMIZER

As previously mentioned, traditional path planning algorithms such as RRT can typically generate a path with less computation time compared to metaheuristic algorithms, but they often sacrifice the quality of the path. However, after a disaster occurs, the need to quickly locate all survivors in a building places higher demands on both the computational time and the quality of the path generated by path planning algorithms. To meet this requirement, similar to references [24]-[26], this study first utilized the RRT algorithm to plan an initial path, and then applied the IMVO algorithm, as designed in reference [26], to further optimize the initial trajectory.

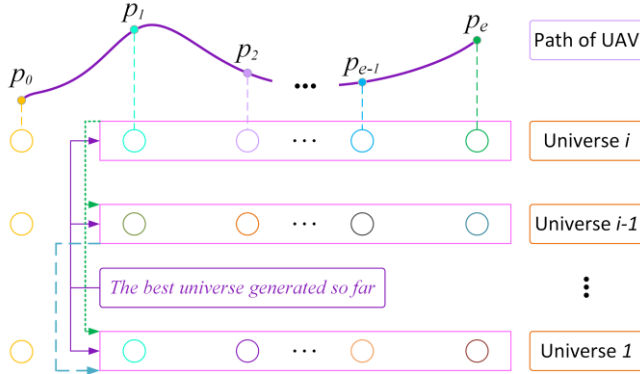


Figure 3. Schematic diagram of optimizing path points based on the IMVO algorithm.

Fig. 3 shows the process of optimizing the initial path using the IMVO algorithm. The first step in this process is to generate the initial universe based on the path points generated by the RRT algorithm. Each universe is represented by a set of path points, excluding the starting and ending points. A set of multiverse includes i universes. The IMVO algorithm, developed in reference [26], relies on three search operations to generate the best quality path. These operations include the roulette wheel-based white hole/black hole transfer operation, the universe-difference operation, and the search strategy of discoverer. It is worth noting that IMVO improves the convergence accuracy of the basic MVO algorithm by utilizing the universe-difference operation and the search strategy of discoverer.

V. SEARCH AND RESCUE EXPERIMENT

In this section, a simulation for UAV-based SAR missions is presented. Before conducting the simulation experiment, we first generated a non-convex building with a height of 36 meters. 17 survivors were randomly generated in the building. Fig. 4 shows the 3D map and the location of survivors in the simulation experiment. To verify the effectiveness of the model designed in this study, the RRT algorithm, the IGSO algorithm [25], and the IMVO algorithm [26] were used in the simulation. Firstly, the RRT algorithm was independently run 50 times, resulting in 50 initial paths. Based on these 50 initial paths, the initial populations of the IGSO and IMVO algorithms were generated, with a population size of 50. Then, the IGSO and IMVO algorithms were each run 50 times.

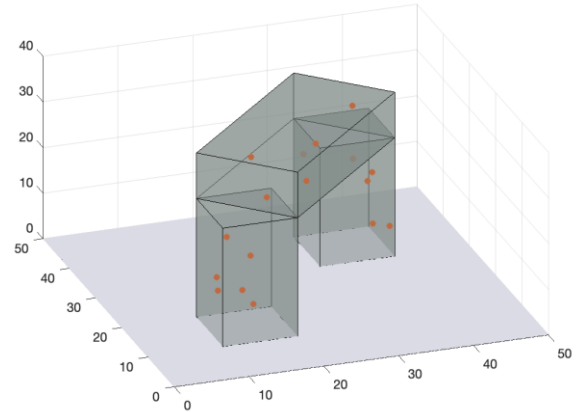


Figure 4. 3D environment for UAV-based search and rescue experiments.

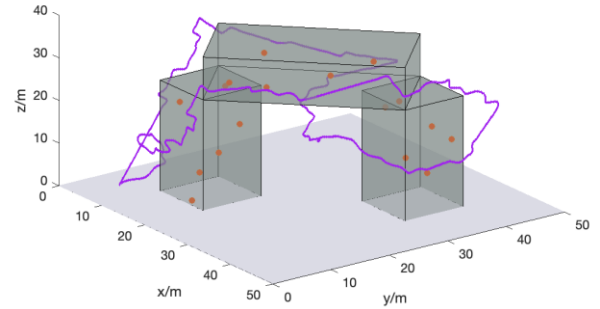


Figure 5. Trajectory planning results for the IMVO algorithm.

Furthermore, Fig. 5 shows the flight trajectory of a randomly selected UAV during 50 runs of the IMVO algorithm. Fig. 6 presents the box plot of the objective function values corresponding to the trajectories generated by the RRT algorithm, the IGSO algorithm, and the IMVO algorithm over 50 runs. It can be observed that the IMVO algorithm designed in reference [26] exhibits the best performance, while the RRT algorithm has the poorest performance. It is worth noting that due to the small number of

buildings in this study and the concentration of survivors, both the IGSO algorithm and the IMVO algorithm were able to successfully located all 17 survivors in all generated trajectories during 50 runs. However, out of the 50 trajectories generated by the RRT algorithm, only 6 were able to successfully located all 17 survivors. Fig.7 shows the average flight time of 50 trajectories generated by three algorithms. The trajectories generated by the IMVO algorithm have the lowest average flight time, while those generated by the RRT algorithm have the longest average flight time.

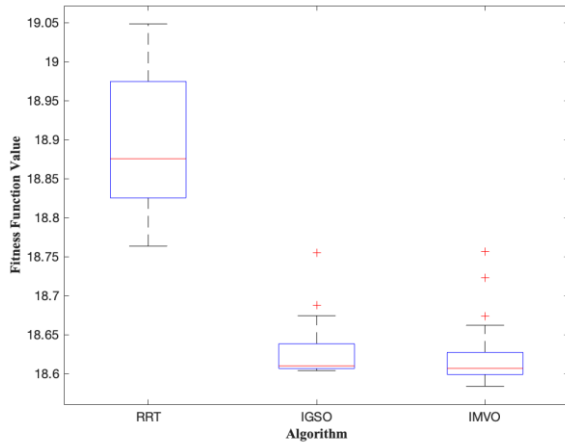


Figure 6. Box plots of fitness functions during 50 runs of three algorithms

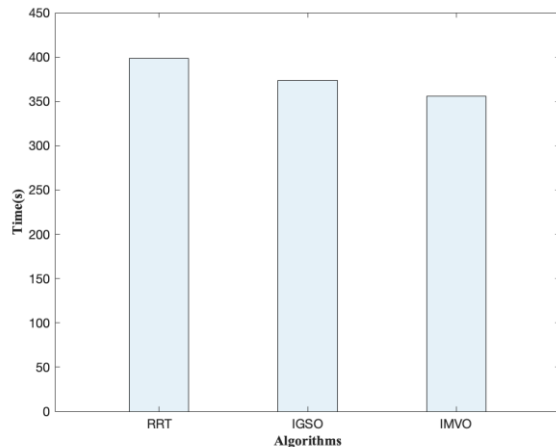


Figure 7. The flight time of UAV trajectories planned by three algorithms

VI. CONCLUSION

According to the requirements of contemporary smart city development for the disaster management, we investigated and proposed an UAV-based IoT solution to automate smart search and rescue operations with optimal path planning in the complex 3D urban disaster environment. We constructed a mathematical model with two objectives, aimed to achieve the UAV search and rescue of the most trapped targets in the shortest time. The RRT, IGSO, and IMVO algorithms were used to solve the above mathematical problem, in order to realize people-oriented SAR activities. Furthermore, we validated our method using a 3D urban disaster environment model and evaluated and compared the effectiveness of the

three trajectory planning algorithms. In future research, we aim to further explore multi-UAV-based SAR path planning in urban environments.

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